SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1005	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
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Date:	09/10/2026	**Duration:**	60 minutes

Code of Conduct

*I ___**KARNAM HARSHITH (192511384)**_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: _____

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. **Understand the Problem Statement**
   - Explain the problem in your own words.
   - Identify the objectives of the proposed system.
   - State the scope of the problem.
2. **Identify Key Stakeholders and Challenges**
   - Identify all the stakeholders involved in the system.
   - Explain the roles and responsibilities of each stakeholder.
   - Discuss the major challenges and issues faced by the stakeholders.
3. **Analyze the Current Approach**
   - Describe the existing system or current process.
   - Identify its strengths and limitations.
   - Analyze the problems or gaps that need to be addressed.
4. **Propose a Suitable Solution Using Software Engineering Principles**
   - Design a software-based solution for the given problem.
   - Describe the major functional modules of the proposed system.
   - Identify the functional and non-functional requirements.
   - Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.
5. **Justify the Chosen Methodology/Framework**

- Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).
- Justify why the selected methodology/framework is suitable for the proposed system.
- Explain how it supports successful project development and maintenance.

6. **Discuss Expected Outcomes and Limitations**
   - Describe the expected benefits of the proposed solution.
   - Explain how the solution addresses the identified challenges.
   - Discuss potential risks, implementation challenges, and limitations.
   - Suggest possible future enhancements.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework	Selects the most suitable SDLC/model/frame	Suitable methodology selected with	Methodology selected but	Outcomes, limitations, or future

Justification (10%)	work and provides strong technical justification.	adequate justification.	justification is weak.	enhancements are missing; report lacks organization and clarity.

Criteria	**Mark**
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25

CO1 – AT1

CASE STUDY ANALYSIS

QUESTION 19

INTELLIGENT IT HELP DESK AND SERVICE MANAGEMENT PLATFORM

Case Study on Intelligent IT Service Management

Submitted By

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Year: 2ND YEAR

Submitted To

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Academic Year 2026–2027

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1. Problem Understanding

1.1 Problem Statement

Organizations depend on IT infrastructure, software applications, computer systems, networks, and other digital services for their daily operations. Employees may frequently face technical problems such as network connectivity issues, software failures, hardware problems, password or account issues, and application errors. These problems need to be reported to the IT support team and resolved within a reasonable time.

In many organizations, employees report IT problems through different channels such as email, phone calls, messages, or direct communication with IT staff. This creates difficulties in systematically recording, categorizing, prioritizing, assigning, and tracking service requests. Important requests may be delayed or overlooked, while IT staff may spend additional time manually managing and following up on tickets.

To overcome these limitations, an **Intelligent IT Help Desk and Service Management Platform** is proposed. The system provides a centralized platform where employees can report IT problems as service tickets. The platform can categorize and prioritize tickets, assign them to appropriate support teams, track their progress, monitor Service-Level Agreements (SLAs), maintain a technical knowledge base, collect user feedback, and generate IT service analytics.

The main purpose of the proposed system is to make IT support **organized, faster, transparent, and easier to manage**.

1.2 Objectives

The main objectives of the Intelligent IT Help Desk and Service Management Platform are:

- To provide a centralized platform for employees to report IT problems.
- To create and maintain IT service tickets with complete problem details.
- To categorize technical problems based on their type, such as network, hardware, software, or account-related issues.
- To prioritize incidents according to their urgency, severity, and impact.
- To automatically or systematically assign tickets to the appropriate IT support team.
- To allow employees and support staff to track the current status of tickets.
- To monitor Service-Level Agreements (SLAs) and identify tickets approaching their deadlines.
- To maintain a technical knowledge base containing solutions for frequently occurring problems.
- To collect feedback from employees after their issues are resolved.
- To generate reports and analytics that help IT managers evaluate service performance.
- To reduce the time required to process and resolve IT service requests.

1.3 Scope of the System

The proposed system focuses on managing the complete lifecycle of IT service requests within an organization. It begins when an employee reports a technical problem and continues until the problem is resolved, feedback is collected, and the ticket is closed.

The system will include **user authentication and role management**, allowing employees, IT support staff, and administrators to access functions according to their roles. Employees can create tickets, provide problem descriptions, view their ticket history, and track the progress of their requests.

IT support staff can view assigned tickets, analyze the reported problem, update ticket status, communicate the resolution, and close completed tickets. The system can categorize and prioritize tickets and route them to the appropriate support team.

The platform will also provide **SLA monitoring, a technical knowledge base, feedback collection, and analytics**. These features will help the organization identify frequently occurring problems, monitor support performance, and improve the overall IT service process.

The major areas covered by the system are:

1. **User and Role Management**
2. **IT Ticket Creation and Management**
3. **Problem Categorization**
4. **Incident Prioritization**
5. **Support Team Assignment**
6. **Ticket Status Tracking**
7. **SLA Monitoring**
8. **Technical Knowledge Base**
9. **User Feedback**
10. **IT Service Analytics and Reporting**

1.4 Problem Characteristics

The problem involves several important characteristics that make a centralized IT service management system necessary.

Fragmented Reporting

When employees use different communication channels to report problems, IT requests can become scattered and difficult to manage.

Manual Processing

Categorizing, prioritizing, and assigning tickets manually increases the workload of IT support staff and may cause delays.

Lack of Visibility

Employees may not know whether their request has been received, assigned, or resolved.

Priority Management

Critical incidents affecting many users need to be handled before minor problems. Without a proper prioritization mechanism, important incidents may not receive immediate attention.

SLA Management

IT service requests may have defined response and resolution deadlines. Manually monitoring these deadlines can result in missed SLA targets.

Repeated Problems

Frequently occurring problems may require repeated support effort when their solutions are not properly documented.

1.5 Need for the Proposed System

A centralized and intelligent help desk system is required to overcome the limitations of traditional IT support methods. By maintaining all service requests in one platform, the organization can obtain better control over the entire ticket lifecycle.

The proposed system can reduce manual effort by assisting with ticket categorization, prioritization, and assignment. It can also improve transparency by allowing employees to track their requests and enabling IT managers to monitor service performance.

The knowledge base can provide solutions for common problems, while analytics can help management identify trends and recurring incidents. Therefore, the proposed platform can contribute to **faster issue resolution, improved IT support efficiency, better SLA compliance, and higher user satisfaction**.

2. Stakeholder & Challenge Analysis

This section is worth **25% of the total marks (6.25/25)**, so it should clearly identify the people involved in the system, their requirements, and the major challenges they may face.

2.1 Stakeholder Identification

A stakeholder is a person, group, or organization that interacts with the system or is affected by its operation. The major stakeholders of the Intelligent IT Help Desk and Service Management Platform are **employees, IT support staff, IT administrators, IT managers, and organizational management**.

The following table summarizes their roles and requirements:

Stakeholder	Role in the System	Major Requirements
Employees / End Users	Report technical problems and track their service requests	Easy ticket creation, ticket tracking, notifications, knowledge-base access, feedback
IT Support Staff	Analyze and resolve reported IT problems	Assigned ticket list, ticket details, priority information, SLA deadlines, status updates
IT Administrator	Manage users, roles, teams, categories and system configuration	User management, role management, team management, system configuration
IT Manager	Monitor IT service performance and support teams	SLA monitoring, performance reports, ticket statistics, resolution-time analysis
Management	Evaluate overall IT service efficiency	Analytics, service reports, recurring problem trends, SLA compliance information

---

2.2 Employee / End User

Employees are the primary users who report technical problems to the IT support team. They may experience issues related to computers, applications, networks, accounts, or other IT services.

The system should allow employees to:

- Create a new IT service ticket.
- Clearly describe their technical problem.
- Select or provide relevant problem information.
- View the generated ticket number.
- Track the status of their request.
- Receive updates regarding their ticket.
- Search the knowledge base for common solutions.
- Provide feedback after the issue is resolved.

A simple and user-friendly interface is important because employees may not have advanced technical knowledge.

2.3 IT Support Staff

IT support staff are responsible for analyzing and resolving the technical problems reported by employees. They need complete information about each assigned ticket to investigate and resolve the issue efficiently.

The system should provide support staff with:

- Assigned ticket information.
- Problem category and priority.
- Employee details.
- Ticket history.
- SLA deadline.
- Ability to update ticket status.
- Ability to record troubleshooting steps and solutions.
- Ability to resolve or escalate tickets.

The system should help support staff focus on resolving problems rather than spending excessive time on manual ticket management.

2.4 IT Administrator

The IT administrator is responsible for managing and configuring the help desk platform.

The administrator should be able to:

- Create and manage user accounts.
- Assign user roles.
- Manage support teams.
- Define ticket categories.
- Configure priority levels.
- Manage SLA rules.
- Maintain knowledge-base content.
- Monitor overall system activity.
- Manage system configuration and access permissions.

Proper role management is necessary to prevent unauthorized users from accessing administrative functions.

2.5 IT Manager

The IT manager is responsible for monitoring the performance of the IT support department.

The system should provide the manager with information such as:

- Number of open and closed tickets.
- Average ticket resolution time.
- Number of overdue tickets.
- SLA compliance percentage.
- Team-wise ticket performance.
- Frequently reported problems.
- User feedback and satisfaction levels.

This information can help the IT manager identify performance problems and improve the efficiency of the support process.

2.6 Organizational Management

Management is interested in the overall effectiveness of IT services rather than individual tickets. They require summarized information that can support decision-making and resource planning.

The system can provide management with:

- Overall IT service performance.
- Ticket volume trends.
- Common categories of incidents.
- Average resolution time.
- SLA compliance.
- Support team performance.
- User satisfaction statistics.

These reports can help management understand whether the organization's IT support services are meeting operational requirements.

2.7 Stakeholder Challenges

Although the proposed platform can improve IT service management, different stakeholders may face several challenges.

Challenge 1 — Accurate Problem Categorization

Employees may describe the same technical problem using different words. For example, one employee may report **"Wi-Fi is not working"**, while another may report **"Unable to connect to office network."**

The system must correctly identify that both problems may belong to the **Network/Connectivity** category.

Challenge 2 — Correct Ticket Prioritization

Different incidents have different levels of urgency and impact. A server failure affecting an entire department should receive higher priority than a minor software issue affecting one employee.

Therefore, the system should consider factors such as:

- Severity
- Urgency
- Business impact
- Number of affected users

when determining ticket priority.

Challenge 3 — Appropriate Team Assignment

A ticket needs to be assigned to the team that has the required expertise.

For example:

```
Network Issue       → Network Support Team
Hardware Problem    → Hardware Support Team
Software Problem    → Application Support Team
Account Problem     → System Administration Team
```

Incorrect assignment can increase resolution time and reduce service efficiency.

Challenge 4 — SLA Monitoring

Service requests may have predefined response and resolution deadlines. If these deadlines are monitored manually, support staff may accidentally miss them.

The proposed system should automatically track SLA deadlines and provide alerts when tickets are approaching or exceeding their defined limits.

Challenge 5 — Managing a Large Number of Tickets

As the organization grows, the number of service requests may increase significantly. The system must be capable of efficiently handling a large number of tickets without affecting performance.

This creates a need for:

- Efficient database management
- Search and filtering
- Proper ticket indexing
- Scalable system architecture

Challenge 6 — Maintaining the Knowledge Base

The knowledge base must contain accurate and up-to-date solutions. Outdated solutions could cause employees or support staff to follow incorrect troubleshooting procedures.

Therefore, knowledge-base articles should be reviewed and updated regularly.

Challenge 7 — Data Security and Access Control

IT service tickets may contain sensitive organizational information. Different users should have different levels of access.

For example:

```
Employee       → Own tickets
Support Staff  → Assigned/authorized tickets
IT Manager     → Service reports
Administrator  → System management
```

Therefore, authentication, authorization, and role-based access control are important challenges.

2.8 Stakeholder–Challenge Relationship

The challenges affect different stakeholders in different ways:

Stakeholder	Major Challenge	Effect
Employees	Difficult ticket tracking	Uncertainty about resolution
IT Support Staff	Large number of manual requests	Increased workload
IT Support Staff	Incorrect categorization/assignment	Longer resolution time
IT Manager	SLA monitoring	Risk of SLA violations
IT Administrator	Security and access control	Risk of unauthorized access
Management	Lack of analytics	Difficulty in making decisions

Summary

The stakeholder analysis shows that each user group has different requirements from the help desk platform. Employees need **simple reporting and transparency**, support staff need **efficient ticket management**, administrators need **system control and security**, while managers and management require **analytics and performance information**.

The major challenges are accurate categorization, prioritization, team assignment, SLA monitoring, scalability, knowledge-base maintenance, and security. Addressing these challenges is essential for developing an effective IT service management platform.

3. Current System Analysis

This section is worth **5 marks**, so you should not only describe the existing system. You need to show **how the current process works, its strengths, weaknesses, and the gaps that justify the proposed system**.

You can copy-paste this directly into your Word document.

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3. Current System Analysis

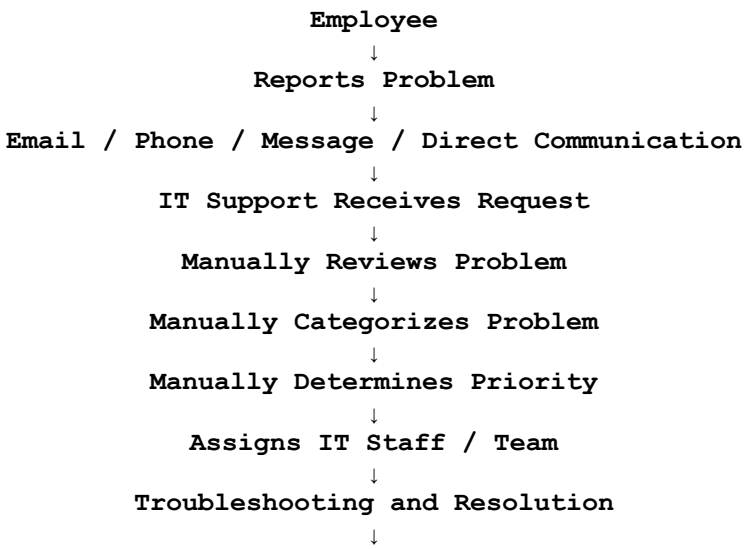
3.1 Overview of the Existing System

In a traditional IT support environment, employees generally report technical problems through different communication channels such as email, telephone calls, messaging applications, or direct communication with IT support staff. The IT support team then manually reviews the request, determines the type and urgency of the problem, assigns it to an appropriate staff member, and communicates with the employee regarding its progress.

Although this approach can work for a small organization with a limited number of users and IT requests, it becomes difficult to manage when the number of employees and service requests increases. The absence of a centralized ticket management system can make it difficult to track requests, monitor deadlines, identify recurring problems, and measure the performance of the IT support team.

3.2 Existing System Workflow

The typical workflow of the current system can be represented as follows:



Communicates Resolution to Employee

↓

Problem Closed

Explanation

First, an employee identifies an IT-related problem and contacts the IT support team through an available communication channel. The support staff receives the request and manually determines its category and priority.

The request is then assigned to an appropriate support staff member or team. The assigned staff member investigates the issue and attempts to resolve it. Once the problem is resolved, the employee is informed and the request is considered complete.

However, because this process is largely manual, maintaining consistent records and monitoring the complete lifecycle of every request can become difficult.

3.3 Strengths of the Existing System

Although the current approach has several limitations, it also has some advantages.

1. Simple Communication

Employees can report problems using familiar communication methods such as email or telephone without learning a new system.

2. Direct Human Interaction

Employees can directly communicate with IT staff and explain complex problems in detail. IT staff can also ask questions and understand the situation through direct communication.

3. Low Initial Cost

A small organization may not need to invest in a dedicated help desk platform because existing communication tools can be used.

4. Suitable for Small Organizations

When the number of employees and IT requests is small, manually handling service requests may still be manageable.

5. Flexible Problem Reporting

Employees can describe unusual or complex problems freely without being restricted by predefined forms or categories.

3.4 Weaknesses of the Existing System

Despite these advantages, the traditional approach creates several problems as the organization grows.

1. Lack of Centralized Ticket Management

Requests received through different channels may not be stored in a single centralized location. This makes it difficult for IT staff to maintain a complete record of all service requests.

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2. Manual Categorization

IT support staff must manually determine whether a request is related to hardware, software, networking, accounts, or another category.

This increases the workload and may result in incorrect categorization.

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3. Manual Prioritization

Without a systematic prioritization mechanism, support staff may find it difficult to identify which incidents require immediate attention.

For example, a network failure affecting an entire department should normally receive higher priority than a minor software issue affecting a single employee.

---

4. Difficult Ticket Tracking

Employees may have limited visibility into the status of their requests.

They may need to contact the IT team repeatedly to ask:

"Has my problem been assigned?"

or

"When will my problem be resolved?"

This creates additional communication overhead.

5. Poor SLA Monitoring

If service deadlines are monitored manually, support staff may miss response or resolution deadlines.

This can result in:

- Delayed service
- SLA violations
- Reduced employee satisfaction

---

6. Repeated Troubleshooting Effort

Common problems may be solved repeatedly because previous solutions are not stored in an organized knowledge base.

For example, if many employees experience the same network configuration problem, IT staff may repeatedly provide the same troubleshooting instructions.

---

7. Limited Performance Analysis

The traditional system makes it difficult to accurately determine:

- Number of tickets received
- Number of resolved tickets
- Average resolution time
- Most common problems
- Team performance
- SLA compliance
- User satisfaction

Without reliable data, management cannot easily evaluate the effectiveness of IT support.

---

8. Higher Risk of Lost or Overlooked Requests

When requests are distributed across emails, phone calls, and messages, an important request may be forgotten or overlooked, especially when the volume of requests increases.

3.5 Gap Analysis

The major gap between the existing system and the required IT service management process is the **lack of a centralized, structured, and trackable ticket management mechanism**.

Existing System	Identified Gap	Required Improvement
Email/phone-based reporting	Requests are scattered	Centralized ticket management
Manual categorization	Time-consuming and inconsistent	System-assisted categorization
Manual priority assignment	Important incidents may be delayed	Structured priority mechanism
Manual team assignment	Incorrect routing possible	Automated/systematic assignment
Manual deadline monitoring	SLA violations may occur	Automated SLA monitoring
No organized knowledge base	Repeated troubleshooting	Centralized knowledge base
Limited reporting	Poor visibility into performance	Analytics dashboard
Limited status visibility	Employees need to contact IT	Real-time ticket tracking

---

3.6 Need for Improvement

The analysis of the current system shows that traditional IT support methods are suitable only when the number of service requests is relatively small. As the organization grows, manual processes become increasingly difficult to manage.

The major areas requiring improvement are **centralized ticket management, automated or systematic categorization, priority management, team assignment, SLA monitoring, knowledge management, status tracking, and analytics**.

Therefore, the existing system should be replaced or enhanced with a centralized IT Help Desk and Service Management Platform that can manage the complete lifecycle of IT service requests.

3.7 Summary of Current System Analysis

The existing IT support process provides simple communication and direct interaction between employees and IT staff. However, its dependence on manual processes creates significant limitations in scalability, tracking, prioritization, SLA management, knowledge management, and performance analysis.

These limitations establish the need for a structured and centralized system. The proposed Intelligent IT Help Desk Platform addresses these gaps by providing **centralized ticket management, intelligent categorization and prioritization, appropriate team assignment, SLA monitoring, knowledge-base support, feedback collection, and analytics**.

4. Proposed Solution Using Software Engineering Principles

This section is worth **3.75 marks**, so the main thing is to clearly explain **what you are proposing and how software engineering principles are applied**.

You can copy-paste this directly into your Word document.

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4. Proposed Solution Using Software Engineering Principles

4.1 Proposed System

To overcome the limitations identified in the existing IT support process, an **Intelligent IT Help Desk and Service Management Platform** is proposed.

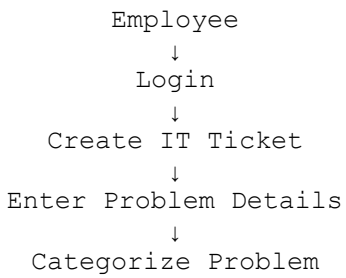
The proposed system will provide a centralized platform for managing IT incidents and service requests throughout their complete lifecycle. Employees can submit their technical problems through the platform, while IT support staff can receive, prioritize, assign, resolve, and close the requests.

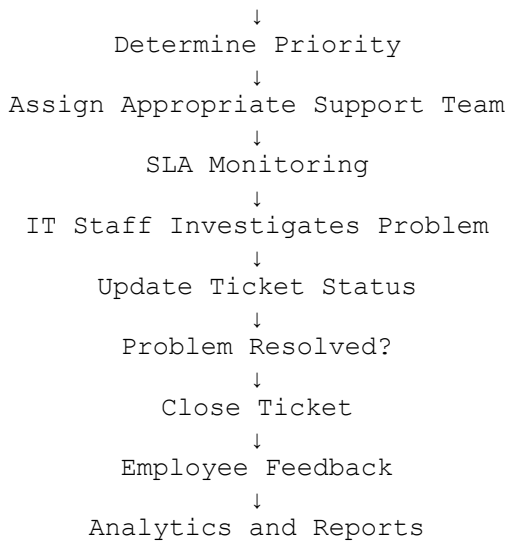
The system will provide intelligent assistance for **ticket categorization and prioritization**. Based on the problem description and relevant information, the system can determine the appropriate category and priority of a ticket. Tickets can then be assigned to the appropriate support team.

The platform will also provide **SLA monitoring, knowledge-base management, notifications, feedback collection, and analytics**. These features will improve the efficiency and transparency of IT service management.

4.2 Proposed System Workflow

The proposed system will follow the following workflow:





The workflow ensures that every service request is recorded and can be tracked from the time it is created until it is resolved and closed.

4.3 Major Components of the Proposed System

4.3.1 User Authentication and Role Management

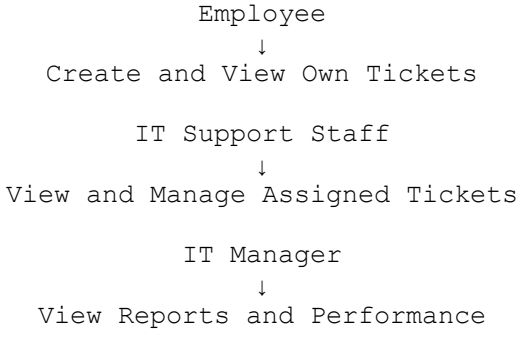
The system will provide secure login and role-based access for different users.

The major roles are:

- Employee
- IT Support Staff
- IT Manager
- Administrator

Each role will have access only to the functions required for its responsibilities.

For example:



```
Administrator
↓
Manage Users, Roles and System Configuration
```

4.3.2 Ticket Management

The ticket management module will be the central component of the platform.

Employees can create a ticket by providing information such as:

- Problem title
- Problem description
- Category
- Relevant attachments/details
- Date and time

Each ticket will receive a unique identification number.

The ticket can then move through different statuses such as:

```
New → Assigned → In Progress → Resolved → Closed
```

If additional investigation is required, the ticket can remain in progress or be escalated to another support team.

4.3.3 Intelligent Categorization

The system can assist in identifying the category of an IT problem based on the ticket description.

For example:

"My laptop cannot connect to the office Wi-Fi."

The system may identify:

```
Category: Network
Subcategory: Connectivity
```

Similarly:

"My computer is not turning on."

may be categorized as:

```
Category: Hardware
Subcategory: System Power
```

This reduces the amount of manual categorization required from IT support staff.

4.3.4 Ticket Prioritization

The system will assign an appropriate priority based on factors such as:

- Severity
- Urgency
- Business impact
- Number of affected users

A possible priority structure is:

Priority	Example
Critical	Organization-wide server/network failure
High	Major department unable to access an important service
Medium	Individual employee unable to use an application
Low	Minor software or configuration issue

This ensures that critical problems receive attention before less urgent requests.

4.3.5 Support Team Assignment

After categorization and prioritization, the ticket can be routed to the appropriate support team.

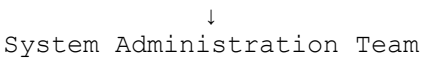
For example:

```
Network Issue
      ↓
Network Support Team

Hardware Issue
      ↓
Hardware Support Team

Software Issue
      ↓
Application Support Team

Account Issue
```



This reduces unnecessary manual routing and helps ensure that tickets reach staff with the required expertise.

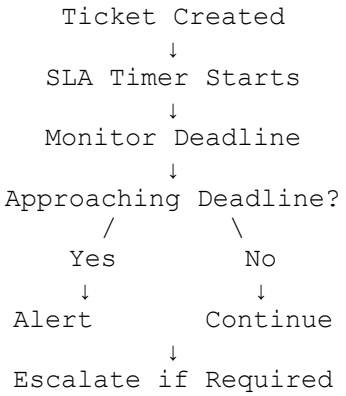
4.3.6 SLA Monitoring

The platform will monitor the response and resolution deadlines associated with tickets.

The system can:

- Record ticket creation time.
- Define response and resolution deadlines.
- Track remaining SLA time.
- Generate alerts for approaching deadlines.
- Identify overdue tickets.
- Support escalation when required.

Example:



4.3.7 Knowledge Base

The knowledge base will contain solutions and troubleshooting instructions for frequently occurring IT problems.

For example:

Problem:

"Unable to connect to Wi-Fi"

↓

Knowledge Base

↓

Possible Solutions:
• Check Wi-Fi is enabled
• Restart network adapter
• Forget and reconnect to network
• Restart the system
• Contact Network Support

This allows employees and support staff to find solutions more quickly and reduces repeated troubleshooting effort.

4.3.8 Feedback and Analytics

After a ticket is resolved, the employee can provide feedback regarding the support received.

The system can collect:

- Satisfaction rating
- Comments
- Resolution feedback

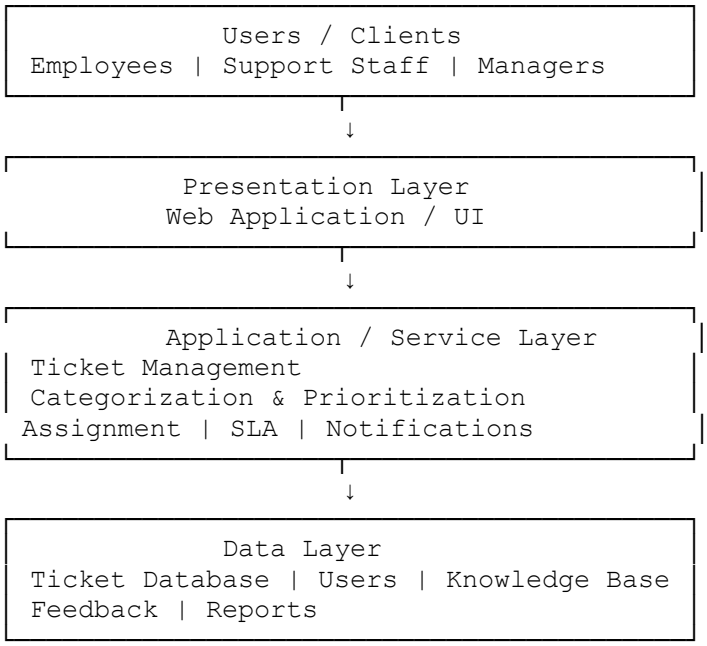
The analytics module can provide information such as:

- Total tickets
- Open tickets
- Resolved tickets
- Average resolution time
- Tickets by category
- SLA compliance
- Team performance
- User satisfaction

This information can help IT managers identify areas that require improvement.

4.4 System Architecture

The proposed platform can be organized into multiple layers:



This layered structure separates different responsibilities and makes the system easier to maintain and modify.

4.5 Application of Software Engineering Principles

The proposed system will follow important software engineering principles to ensure that it is reliable, maintainable, secure, and scalable.

4.5.1 Modularity

The system will be divided into separate modules such as authentication, ticket management, SLA management, knowledge base, and analytics.

This allows each module to be developed and tested independently.

4.5.2 Maintainability

The system should use a structured architecture and well-organized code so that future changes can be made without significantly affecting existing functionality.

For example, a new ticket category can be added without redesigning the complete system.

4.5.3 Scalability

The system should be capable of supporting an increasing number of users and tickets.

A scalable architecture allows the organization to expand the platform as the number of employees and IT requests increases.

4.5.4 Security

Since the system contains organizational and employee information, security is essential.

The proposed system should use:

- Authentication
- Authorization
- Role-Based Access Control
- Input validation
- Secure database access
- Appropriate password protection

Users should only be allowed to access information relevant to their roles.

4.5.5 Usability

The platform should provide a simple and intuitive interface. Employees should be able to create and track tickets without requiring advanced technical knowledge.

Clear forms, understandable status indicators, and simple navigation can improve usability.

4.5.6 Reliability

The system should maintain accurate ticket information and ensure that service requests are not lost during normal operation.

Database transactions, validation, error handling, and appropriate backup mechanisms can improve system reliability.

4.5.7 Reusability

Common components such as authentication, notification services, validation functions, and reporting components should be designed for reuse across different parts of the system.

This reduces duplicate development effort and improves consistency.

4.6 Benefits of the Proposed Solution

The proposed platform can provide the following improvements:

- Centralized management of IT service requests.
- Faster ticket categorization and assignment.
- Better prioritization of critical incidents.
- Improved visibility of ticket status.
- Better SLA monitoring.
- Reduced repetitive troubleshooting.
- Improved communication between employees and IT support.
- Better IT service performance analysis.
- Improved user satisfaction.
- More organized and efficient IT support operations.

4.7 Summary

The proposed Intelligent IT Help Desk and Service Management Platform addresses the major weaknesses identified in the existing system by introducing centralized ticket management, intelligent categorization and prioritization, appropriate team assignment, SLA monitoring, knowledge management, feedback, and analytics.

The use of software engineering principles such as **modularity, maintainability, scalability, security, usability, reliability, and reusability** ensures that the proposed system can be developed as a structured, maintainable, and reliable software solution.

5. Methodology / Framework Justification

This section is worth **2.5 marks**, so keep it focused but properly justify **why the selected methodology is suitable for this project**.

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5. Methodology / Framework Justification

5.1 Selected Methodology – Agile Scrum

The proposed Intelligent IT Help Desk and Service Management Platform will follow the **Agile Scrum methodology**.

Agile Scrum is an iterative and incremental software development methodology in which the system is developed through a series of short development cycles called **Sprints**. Each Sprint focuses on developing, testing, and improving a specific set of system features.

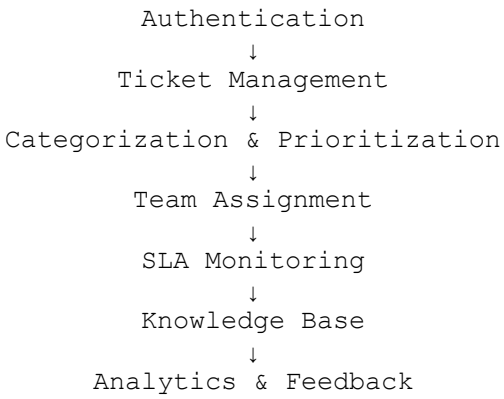
Scrum is suitable for this project because the requirements of an IT help desk system may change based on feedback from employees, IT support staff, and managers. Developing the system incrementally allows the team to obtain feedback at an early stage and make improvements during subsequent Sprints.

5.2 Why Agile Scrum is Suitable

1. Incremental Development

The complete system can be divided into smaller features and developed step by step.

For example:



Instead of waiting until the entire system is completed, usable features can be developed during individual Sprints.

---

2. Changing Requirements

The requirements of an IT help desk may change after employees and IT staff interact with the system.

For example, users may request:

- Additional ticket categories
- New notification options
- Improved ticket filtering
- Additional analytics
- Changes to the knowledge base

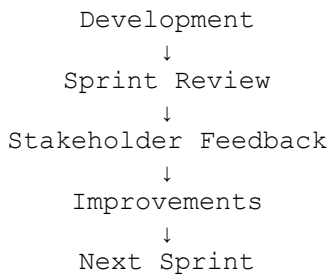
Agile allows such requirements to be incorporated into future Sprints.

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3. Continuous Feedback

At the end of each Sprint, the developed features can be demonstrated to stakeholders.

Their feedback can then be used to improve the next version of the system.



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4. Continuous Testing

Testing can be performed throughout development instead of only at the end of the project.

Each completed module can be tested before integrating it with other modules.

This helps identify defects early and reduces the cost and effort of fixing problems later.

---

5. Early Delivery of Important Features

The most important functionality can be developed first.

For example, the first Sprint can provide basic login and ticket creation functionality. This allows stakeholders to evaluate the basic workflow before advanced features are developed.

5.3 Proposed Scrum Sprint Plan

The project can be divided into the following Sprints:

Sprint	Major Activities	Expected Output
Sprint 1	Requirement analysis, user roles, authentication	Login and role management
Sprint 2	Ticket creation, viewing and status management	Basic ticket management
Sprint 3	Categorization, prioritization and team assignment	Intelligent ticket routing
Sprint 4	SLA monitoring and notifications	SLA tracking system
Sprint 5	Knowledge base and feedback	Knowledge and feedback modules
Sprint 6	Analytics, integration and system testing	Complete integrated system

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5.4 Scrum Roles

The project team can use the following Scrum roles:

Product Owner

The Product Owner represents the requirements of the organization and prioritizes the product backlog.

Scrum Master

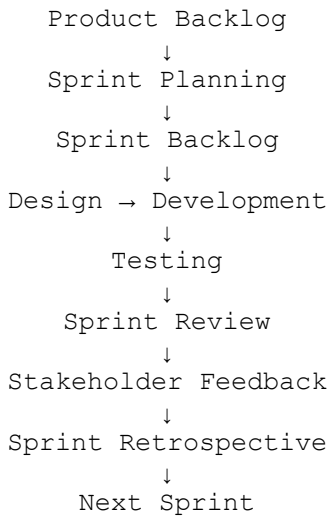
The Scrum Master facilitates the Scrum process, helps remove development obstacles, and ensures that the team follows Scrum practices.

Development Team

The development team is responsible for designing, developing, testing, and integrating the different components of the help desk platform.

5.5 Scrum Development Cycle

The development process can be represented as:



The cycle is repeated until the required features of the system are completed.

5.6 Advantages of Using Agile Scrum

Using Agile Scrum for this project provides several advantages:

- Supports changing requirements.
- Encourages continuous stakeholder feedback.
- Allows incremental development.
- Enables continuous testing.
- Helps identify problems early.
- Allows important features to be developed first.
- Improves communication within the development team.
- Reduces the risk associated with developing the complete system at once.

5.7 Methodology Justification

Agile Scrum is appropriate for the proposed IT Help Desk Platform because the project contains multiple independent but interconnected modules, and the requirements may evolve based on user feedback. The system can therefore be developed incrementally, with each Sprint delivering a functional set of features.

For example, basic ticket management can be developed first, followed by intelligent categorization, SLA monitoring, knowledge management, and analytics. Continuous testing and stakeholder feedback throughout the development process can improve the quality and usability of the final system.

Therefore, **Agile Scrum provides a flexible and systematic approach for developing the proposed IT Help Desk and Service Management Platform.**

6. Expected Outcomes and Limitations

This section completes the case study by explaining **what improvements are expected from the proposed system** and **what limitations may still exist**. It is important to include both because a good software engineering analysis should not claim that the proposed solution completely eliminates every problem.

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6. Expected Outcomes and Limitations

6.1 Expected Outcomes

The proposed Intelligent IT Help Desk and Service Management Platform is expected to improve the efficiency, transparency, and reliability of IT support operations. By centralizing service requests and introducing structured ticket management, the organization can manage IT incidents more effectively.

1. Faster Ticket Processing

The system can reduce the time required to process IT service requests by providing structured ticket creation, categorization, prioritization, and team assignment.

Instead of manually searching through emails or messages, support staff can access all relevant ticket information from a centralized platform.

---

2. Improved Ticket Prioritization

The proposed system can classify tickets according to their urgency, severity, and impact.

Critical incidents affecting a large number of employees can therefore receive higher priority than minor individual issues.

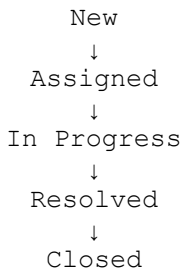
This helps IT staff focus their efforts on the problems that have the greatest effect on organizational operations.

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3. Better Ticket Tracking and Transparency

Employees can view the current status of their service requests without repeatedly contacting IT support.

A ticket can move through stages such as:



This provides greater transparency for both employees and IT support staff.

---

4. Improved SLA Compliance

The system can continuously monitor response and resolution deadlines.

When a ticket approaches its SLA deadline, the system can generate an alert. Overdue tickets can also be identified and escalated when required.

This can help the organization improve its SLA compliance.

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5. Reduced Repetitive Troubleshooting

The knowledge base can store solutions for frequently occurring technical problems.

Employees and support staff can search for existing solutions before spending significant time investigating a repeated problem.

This can reduce repetitive work and improve support efficiency.

---

6. Improved IT Support Productivity

By reducing manual activities such as ticket tracking, categorization, assignment, and reporting, IT support staff can spend more time on actual problem resolution.

The centralized platform can also help staff organize their workload according to ticket priority and SLA deadlines.

---

7. Better Management Decision-Making

The analytics module can provide useful information about IT service performance.

For example, management can analyze:

- Number of tickets received
- Number of resolved tickets
- Average resolution time
- Frequently occurring problems
- Tickets by category
- SLA compliance
- Team performance
- User satisfaction

This information can support better planning and resource allocation.

---

8. Improved User Satisfaction

Employees can benefit from faster response, better communication, ticket tracking, and access to troubleshooting information.

Feedback collected after ticket resolution can also help the organization identify areas where IT support needs improvement.

---

9. Centralized IT Service Records

All service requests, updates, resolutions, and feedback can be maintained in one centralized system.

This creates a historical record that can be useful for:

- Future troubleshooting
- Performance analysis
- Auditing
- Identifying recurring incidents
- Improving IT service processes

6.2 Key Performance Indicators

The success of the proposed system can be evaluated using measurable indicators.

KPI	Purpose
Average Resolution Time	Measures how quickly tickets are resolved
Average Response Time	Measures how quickly support staff respond
SLA Compliance Rate	Measures the percentage of tickets resolved within SLA
Ticket Resolution Rate	Measures the number of tickets successfully resolved
User Satisfaction Score	Measures employee satisfaction with IT support
Reopened Ticket Rate	Identifies tickets that were not properly resolved initially
Ticket Volume by Category	Identifies frequently occurring IT problems

These indicators can help IT managers determine whether the new system is improving service quality.

6.3 Limitations of the Proposed System

Although the proposed system provides several benefits, it also has certain limitations.

1. Incorrect Categorization

The intelligent categorization mechanism may incorrectly classify a ticket if the employee provides incomplete, unclear, or ambiguous information.

For example, a vague description such as:

"Internet problem"

may not provide enough information to determine the exact cause or category.

Therefore, human verification may still be required for some tickets.

---

2. Dependency on Data Quality

The accuracy of analytics, prioritization, and classification depends on the quality of the information stored in the system.

Incomplete or incorrect ticket information can affect the reliability of system-generated results.

3. Knowledge Base Maintenance

The knowledge base must be regularly reviewed and updated.

If troubleshooting information becomes outdated, employees or support staff may receive incorrect or ineffective solutions.

---

4. Initial Development and Setup Effort

Developing and configuring the platform requires time and resources.

The organization must initially define:

- User roles
- Ticket categories
- Priority rules
- SLA policies
- Support teams
- Knowledge-base content

---

5. User Adoption

Employees and IT staff must be willing to use the new platform consistently.

If employees continue to report problems through informal channels instead of creating tickets, the organization may not receive the full benefits of centralized ticket management.

---

6. Security Risks

The platform may contain sensitive information about employees, systems, applications, and organizational IT infrastructure.

Therefore, inadequate security controls could expose the system to unauthorized access or data leakage.

Strong authentication, authorization, access control, and secure data management are required.

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7. Integration Challenges

If the organization already uses other IT management systems, integrating the proposed platform with existing applications may require additional development and configuration.

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8. Scalability Requirements

A basic prototype may work effectively for a limited number of users and tickets. However, an enterprise-level deployment with a very large number of users and service requests may require additional infrastructure, database optimization, monitoring, and architectural improvements.

6.4 Future Enhancements

The proposed system can be further improved in the future by introducing additional capabilities such as:

- AI-powered conversational chatbot support.
- Advanced natural-language ticket classification.
- Predictive identification of recurring incidents.
- Automated resolution recommendations.
- Mobile application support.
- Integration with email and organizational communication systems.
- Integration with existing enterprise IT service management tools.
- Advanced predictive analytics.
- Automated escalation and notification mechanisms.

These enhancements can be considered after the basic ticket management platform has been successfully implemented and evaluated.

6.5 Overall Conclusion

The Intelligent IT Help Desk and Service Management Platform provides a structured approach to managing IT service requests. By centralizing ticket management, supporting categorization and prioritization, assigning requests to appropriate teams, monitoring SLAs, maintaining a knowledge base, and providing analytics, the system can improve the efficiency and transparency of IT support operations.

However, successful implementation depends on factors such as **data quality, user adoption, knowledge-base maintenance, security, and appropriate system configuration**. Therefore, the

proposed platform should be continuously evaluated and improved based on user feedback and service performance.